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SideQuest: A Social Chatting and Group Hangout Decision-Making Application

Project Progress Report

Introduction

SideQuest is being developed as a social chatting application that allows users to communicate through private and group conversations, with its primary feature, “Start a SideQuest,” enabling members of a group chat to collaboratively plan hangouts and make decisions. During this stage, our main objective was to establish the basic structure of the application and develop an interactive frontend prototype, along with an initial backend foundation for user authentication.

Selection of Development Framework and Programming Language

We selected Flutter as the primary framework for developing the SideQuest user interface. One of the main reasons for choosing Flutter was its cross-platform development capability, allowing us to develop an application that can eventually run on both Android and iOS without maintaining completely separate implementations. Since Flutter uses Dart as its primary programming language, we selected Dart for implementing the application's frontend functionality, including form validation, screen transitions, navigation selection, and the display of sample conversations.

Development Approach and Implementation

The Flutter development environment was set up using Visual Studio Code (VS Code), and Google Chrome was used as the target platform for testing interface layouts, navigation, and user interactions. We organized the application into separate Dart files: **main.dart**, **splash_screen.dart**, **auth_screen.dart** and **home_screen.dart**. A dark-themed interface with purple accents was selected to establish a consistent visual identity for SideQuest.

Frontend Screens and Implemented Features

1. Splash Screen: The splash screen contains the SideQuest application name, a temporary rocket-shaped logo, and the tagline “Less deciding. More doing.” It remains visible for approximately two seconds before automatically navigating to the login interface.

2. Login and Sign-Up Screens: The login interface contains fields for entering an email address and password, and the sign-up interface additionally includes a field for the user's name. Basic frontend validation ensures that required fields are completed, email addresses follow a valid format, and passwords contain at least six characters. At this stage, the login and registration processes are simulated.

3. Home Screen and Messaging Interface: The home screen contains three main navigation sections: Chats, Groups and Profile. It uses hardcoded sample conversations, and the chat screen allows messages to be displayed locally during the current session. An important design decision was to keep the “Start a SideQuest” feature exclusively within group conversations. Selecting this button opens the initial SideQuest planning interface, where users can choose activities, select a date, and enter an estimated budget.

Current Application Workflow

Splash Screen → Login / Sign Up → Home Screen → Private Chat / Group Chat
Group Chat → Start a SideQuest → Select Activities → Choose Date → Enter Budget → Display Planning Summary

Backend Development and Authentication

For the backend, we selected FastAPI with Python instead of using Firebase as a managed backend platform. This gives us direct control over our API endpoints, application logic, authentication flow, and future WebSocket-based communication. The stack also includes PostgreSQL, SQLAlchemy 2.x, Pydantic, pwdlib with Argon2, PyJWT and Uvicorn. A SQLAlchemy User model has been defined for the users table, with unique constraints on username and email.

- **Password Hashing:** During registration, the backend hashes the submitted password using Argon2 and stores the resulting hash in PostgreSQL. During login, it verifies the submitted password against the stored hash instead of retrieving or decrypting the original password.
- **JWT Authentication:** After a successful login, SideQuest issues a JSON Web Token (JWT) containing the user's ID and an expiration time, signed using the HS256 algorithm. The current access-token lifetime is 30 minutes.
- **Implemented APIs:** POST /auth/register, POST /auth/login, GET /auth/me, GET /db-check and GET /.
- **Challenges:** During development, we encountered configuration and integration issues involving environment variables, JWT authorization in Swagger UI, and API/database testing. We are addressing these issues by testing endpoints individually through Swagger UI and inspecting server output and HTTP status codes.

Current Progress and Future Development

The next major integration task is to connect the Flutter frontend to the FastAPI authentication endpoints and verify the complete registration and login flow using real database records. Our next development phase will also focus on developing more detailed group preferences, activity suggestions, collaborative voting interfaces, and a finalized hangout-plan screen.

Conclusion

The current development progress of SideQuest includes a Flutter/Dart frontend prototype and an initial FastAPI backend foundation for user authentication. Most importantly, the integration of the Start a SideQuest option directly within group chats reflects the primary concept of our proposed application: combining ordinary social communication with structured group hangout planning.